Recommendations — pancreatic-mets-kras-g12d-basal-arid1a-k9r3

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** pancreas
- **Histology:** pancreatic ductal adenocarcinoma (PDAC); PurIST RNA signature classifies the tumor as the basal-like transcriptional subtype
- **Stage:** IV (metastatic): large primary pancreatic mass with multiple liver metastases each >1 cm
- **ECOG:** 1
- **Age band:** 70-79
- **Sex:** male
- **Biomarkers:** KRAS G12D mutated; MLH1 mutation somatic mutation reported (single MMR-gene variant; tumor is microsatellite stable and germline-negative); SMAD4 E520 *stop-gain (predicted loss of function); TP53 Y126S and A138V (two missense variants); ARID1A frameshift (predicted loss of function); TMB 8.6 mutations/Mb; Microsatellite status MSS (microsatellite stable); PurIST subtype basal-like; Germline status no germline variants detected; Gene rearrangements none detected; HLA type A*01:01, A*02:01, B*37:01, B*44:02, C* 05:01, C*06:02; Additional xR variants ABCC3 missense; ASXL1 frameshift; BCL7A splice-region; BCLAF1 stop-gain; LRP1B splice-region
- **Efficacy/toxicity weight:** 0.5

7 ranked therapeutic options across 2 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Kras G12D interventions

1. daraxonrasib (RMC-6236), pan-RAS(ON) inhibitor — front-line via RASolute 303 (NCT07491445)

Likelihood of effect: High in 2L RAS G12 PDAC — OS HR 0.40 (95% CI 0.30-0.53), RASolute 302 (n=500); first-line benefit inferred from the unread RASolute 303.

Toxicity burden: Moderate (rash G3+ 14%, stomatitis G3+ 12%; G3+ TRAE 43.6% overall, below chemo)

Counter-productive MoA:Low (Mucocutaneous toxicity can force dose reductions that erode the exposure the survival signal depends on; no mechanism-level antagonism)

Overall:The only feature-targeting option with randomized phase 3 survival evidence; first-line benefit inferred from an unread trial and the allele estimate borrowed from a RAS G12 subgroup.

Key references: PMID 42223072 · NCT07491445 · PMID 38593348

2. GFH375 / VS-7375, KRAS G12D(ON/OFF)-selective inhibitor + cetuximab — 1L cohort via NCT07644559

Likelihood of effect: High allele-matched response — ORR 41% (90% CI 30-52), ESMO 2025 LBA84 (n=59), but single-country phase 1/2 and one line beyond the treatment-naive patient.

Toxicity burden: Moderate (neutropenia G3+ 8%, anemia G3+ 8%, diarrhea G3+ 3%; cetuximab rash/hypomagnesemia uncharacterized in combination)

Counter-productive MoA:Low (Cytopenias and cetuximab dermatologic toxicity can force dose reductions; no mechanism-level antagonism of the therapeutic goal)

Overall:The highest allele-matched single-agent response on the board with an open 1L slot, but the read is China-based, one line early, and the cetuximab doublet's tolerability at 79 is uncharacterized.

Key references: NCT07644559 · doi:10.1016/j.annonc.2025.09.099 · NCT07262567

3. zoldonrasib (RMC-9805), KRAS G12D(ON)-selective inhibitor — front-line via RASolute 305 (NCT07621718)

Likelihood of effect: Moderate, allele-matched — ORR 30% / DCR 80% in G12D PDAC (phase 1, n=40, no CI; PFS/DoR immature). RASolute 305 will test it front-line.

Toxicity burden: Low (nausea 27%, diarrhea 20%, rash 10%; no G4/5 events at monotherapy dose)

Counter-productive MoA:Moderate (Critic and concensusite dissented on the thin single-arm base; durability unmeasured, so on-target activity may not translate to a durable effect)

Overall:The allele-matched, gentlest agent in the set — best tolerability for a 79-year-old, but the evidence is a single-arm n=40 abstract with no durability readout.

Key references: NCT07621718 · doi:10.1200/JCO.2025.43.4_suppl.724 · NCT06040541

4. INCB161734, KRAS G12D-selective inhibitor + chemotherapy — front-line via DAWN-303 (NCT07522073)

Likelihood of effect: Moderate to high single-agent — ORR 37% / DCR 78% in pretreated G12D PDAC (phase 1, n=41, no CI; PFS not reached at ~5 mo). DAWN-303 tests it front-line.

Toxicity burden: High (combination-cohort G3+ neutropenia ~40%; monotherapy GI 50-60% grade 1-2)

Counter-productive MoA:Moderate (Conservative dissented on additive marrow toxicity: chemo-combination neutropenia can force dose reductions that erode delivered intensity at 79)

Overall:Strongest single-agent response number in the dossier, but the enrollable front-line construct bundles chemotherapy with ~40% grade >=3 neutropenia and no dedicated combination-safety data.

Key references: NCT07522073 · doi:10.1200/JCO.2026.44.2_suppl.654 · NCT06179160

5. HRS-4642 (KRAS G12D inhibitor) + gemcitabine/nab-paclitaxel — front-line via NCT07232875

Likelihood of effect: High front-line response but confounded — ORR 63.3% (95% CI 43.9-80.1, n=30) on a chemo backbone; monotherapy contribution only ~20.8%.

Toxicity burden: High (grade >=3 TRAE 90.3%, mostly hematologic; one treatment-related death; hypertriglyceridemia/hypercholesterolemia signature)

Counter-productive MoA:Moderate (Backbone myelosuppression plus one treatment-related death can force intensity cuts that erode delivered benefit in an older marrow-limited patient)

Overall:The highest front-line ORR on the exact allele, but a 90.3% grade >=3 rate with a treatment-related death, an IV backbone, and a not-yet-recruiting phase 3 down-rank it against a toxicity-averse 79-year-old.

Key references: NCT07232875 · doi:10.1038/s41591-026-04538-9 · doi:10.1016/j.annonc.2025.08.1484

6. Pan-KRAS(ON) first-in-human inhibitors (BBO-11818, BLU-924, PF-07934040, LY3962673) — phase 1 baskets covering G12D

Likelihood of effect: Low / unquantified — no efficacy denominator; the only public signal is a single BBO-11818 PDAC partial response with no ORR (registry-stage class).

Toxicity burden: Low (none characterized — early reads describe predominantly grade 1-2 GI, no per-term G3+ table published)

Counter-productive MoA:Low (Pre-data mechanism with no plausible counter-productive vector established; the risk is absence of evidence, not a mechanistic antagonism)

Overall:On-axis pan-KRAS agents covering G12D with recruiting phase 1 slots, but no efficacy denominator exists yet, so they are a bank-for-later class rather than a present recommendation.

Key references: NCT06917079 · NCT07629960 · NCT06586515

Evidence in detail

daraxonrasib (RMC-6236), pan-RAS(ON) inhibitor — front-line

Wolpin et al. *N Engl J Med* 2026

- **Disease context:previously treated metastatic RAS-mutant pancreatic ductal adenocarcinoma** (2L+) — metastatic PDAC after one prior line of chemotherapy; ~92% RAS G12, KRAS G12D included in the G12 stratum
- **n:** 500
- **Toxicities:any treatment-related AE** (grade >=3): 43.6%; **rash** (grade >=3): 14%; **stomatitis** (grade >=3): 12%; **treatment discontinuation due to TRAE** (grade any): 1.2%
- **Efficacy:OS** 13.2 months; **HR_OS** 0.4 (95% CI 0.3-0.53); **HR_OS** 0.4 (95% CI 0.3-0.54); **OS_rate** 53.2%; **PFS** 7.2 months; **HR_PFS** 0.49; **ORR** 31.6%; **AE_rate** 43.6%; **discontinuation_rate** 1.2%

Arid1A Loss interventions

1. ceralasertib (AZD6738), ATR inhibitor (ARID1A synthetic-lethal axis) — via NCT03682289, gated on BAF250a IHC loss

Likelihood of effect: Low for PDAC — confirmed ORR 14% across ARID1A-deficient tumors (PMID 41686845), activity concentrated in gynecologic histologies with zero pancreatic responders shown.

Toxicity burden: Moderate (ATR-class anemia / thrombocytopenia / neutropenia; per-term G3+ rates unreported, zero treatment-related deaths)

Counter-productive MoA:Moderate (Cross-tumor extrapolation with no PDAC responder; the synthetic-lethal effect may not engage basal-like PDAC, and class anemia limits exposure at 79)

Overall:A mechanistically independent banked second axis off the ARID1A frameshift, but no PDAC responder exists, entry is BAF250a-IHC-gated, and the lead basket is not recruiting.

Key references: NCT03682289 · PMID 41686845 · PMID 27958275

Evidence in detail

ceralasertib (AZD6738), ATR inhibitor (ARID1A synthetic-lethal axis) —

Zhu et al. *Clin Cancer Res* 2026

- **Disease context:ARID1A-deficient advanced solid tumors (gynecologic-predominant)** (2L+) — 29 evaluable, ARID1A loss by IHC required; mostly gynecologic (endometrioid endometrial, ovarian clear cell); non-gynecologic tumors under-represented
- **n:** 29
- **Toxicities:treatment-related death** (grade 5): 0%
- **Efficacy:ORR** 14%; **ORR** 31%; **DoR** 33.7 months